



Event Hubs – Retaining Data

1. In this lab we will see how we can retain data using Event hubs.
2. Now from our event hub namespace we will create a new event hub.

Create Event Hub ...

Event Hubs

Basics Capture Review + create

Event Hub Details

Enter required settings for this event hub, including partition count and message retention.

Name * ⓘ

loghub

Partition count ⓘ



2

Retention

Configure retention settings for this Event Hub. [Learn more](#)

Cleanup policy ⓘ

Delete

Retention time (hrs) * ⓘ

24

min. 1 hour, max. 168 hours (7days)

3. So, basically Event hub is a place where you cannot persist the data but if you want to do that then you can make use of the capture and persist your data in a storage account.
4. Turn on the Capture feature and then in the capture provider choose Azure Storage Account and click on Select Container to choose the container.
5. Now what you can do is if you don't have a storage account first go **create one** and in that, you also need to create a container. When both of these steps are completed then come back to the event hub namespace and create an event hub.

Capture Details

Azure Event Hubs Capture enables you to automatically deliver the streaming data in Event Hubs to an Azure Blob storage or Azure Data Lake Store account of your choice, with the added flexibility of specifying a time or size interval. Setting up Capture is fast, there are no administrative costs to run it, and it scales automatically with Event Hubs throughput units. Event Hubs Capture is the easiest way to load streaming data into Azure, and enables you to focus on data processing rather than on data capture. [Learn more about capture.](#)

Capture On Off

Note: Enabling Capture will result in additional charges to this account. [Learn more about our pricing.](#)

Time window (minutes) 5

Size window (MB) 300

☐ Do not emit empty files when no events occur during the Capture time window

Capture Provider Azure Storage Account

Azure Storage Container * eventhub [Select Container](#)

- After that move to the review page and create your event hub.
- Once your event hub is created now, we want to send some logs on our event hub.
- But this time we will use a **Web App** to send logs to the event hub. So, you have to create a Web app.
- Open your web app search for diagnostic settings and click on Add.

newwebapp121 | Diagnostic settings ✎ ☆ ...

Web App

diago x x << Refresh Feedback

Diagnose and solve problems

Monitoring

Diagnostic settings

Support + troubleshooting

Support + Troubleshooting

Diagnostic settings are used to configure streaming export of platform logs and metrics for a resource to the destination of your choice. You may create up to five different diagnostic settings to send different logs and metrics to independent destinations. [Learn more about diagnostic settings](#)

Name	Storage account	Event hub	Log Analytics worksp...	Partner solution	Edit setting
No diagnostic settings defined					
+ Add diagnostic setting					

Click 'Add Diagnostic setting' above to configure the collection of the following data:

- HTTP logs
- App Service Console Logs
- App Service Application Logs
- Access Audit Logs
- IPSecurity Audit logs
- App Service Platform logs
- App Service Authentication logs (preview)
- AllMetrics

- Then you have to choose HTTP logs in the categories and stream them to an event hub. Just remember to choose your event name. Click on save.

Diagnostic setting ...

Save Discard Delete Feedback

[more about the different log categories and contents of those logs](#)

Diagnostic setting name * ✓

Logs

Categories

☒ HTTP logs

☐ App Service Console Logs

☐ App Service Application Logs

☐ Access Audit Logs

☐ IPSecurity Audit logs

☐ App Service Platform logs

☐ App Service Authentication logs (preview)

Metrics

☐ AllMetrics

Destination details

☐ Send to Log Analytics workspace

☐ Archive to a storage account

☒ Stream to an event hub

For potential partner integrations, [click to learn more about event hub integration](#).

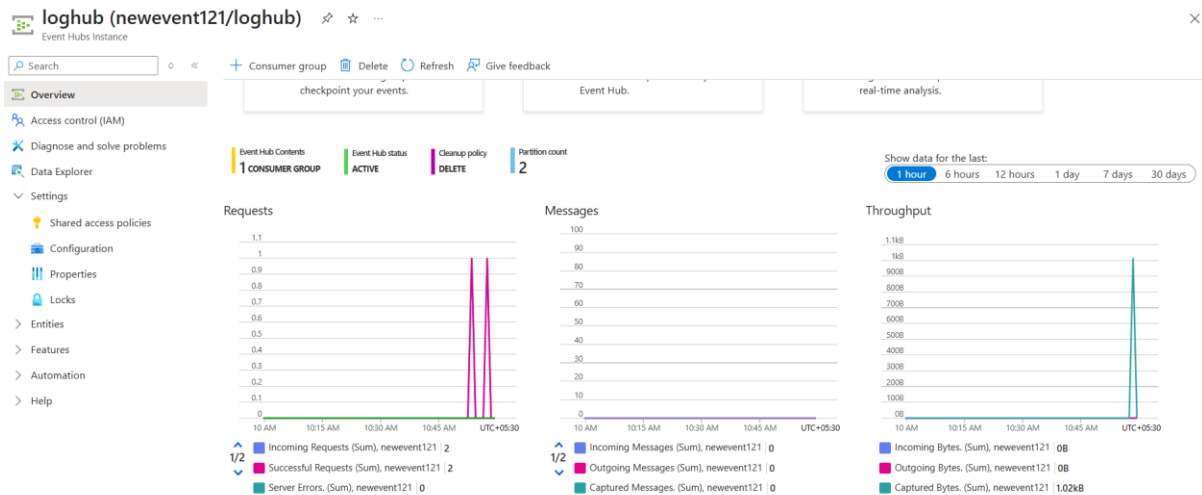
Subscription

Event hub namespace *

Event hub name (optional) ⓘ

Event hub policy name

11. After that you need to copy the DNS name of your Web app and in a new tab just run it and refresh it multiple times to get the logs in the event hub.
12. If you go to the event hub and scroll down on the overview page you will see that the requests are coming in.



13. Then just refresh your web app multiple times and wait for 5 minutes for the logs to reach inside your storage account.
14. After some time if you go to your storage account, and go inside your container you will see that a file has been stored there.

eventhub

Container

Search

Upload

Change access level

Refresh

Delete

Change tier

Acquire lease

Break lease

View snapshots

Create snapshot

Give feedback

Overview

Diagnose and solve problems

Access Control (IAM)

Settings

Authentication method: Access key (Switch to Microsoft Entra user account)

Location: eventhub / newevent121 / loghub / 1 / 2025 / 03 / 12 / 05 / 21

Search blobs by prefix (case-sensitive)

Show deleted blobs

Add filter

Name	Modified	Access tier	Archive status	Blob type	Size	Lease state	
☐ [-]							...
☐ 59.avro	3/12/2025, 10:56:59 ...	Hot (Inferred)		Block blob	508 B	Available	...

15. Now in the end just come back to the event hub and here in the Shared access policies create a listen policy. Then copy the connection string of your policy.

loghub (newevent121/loghub)

Shared access policies

☆ ...

Event Hubs Instance

Search

+

Add

Overview

Access control (IAM)

Diagnose and solve problems

Data Explorer

Settings

Shared access policies

Configuration

Properties

Search to filter items...

Policy	Claims
ListenPolicy	Listen

SAS Policy: ListenPolicy



Event Hub



Regenerate primary key



Regenerate secondary key



Delete

Event Hubs access control with Shared Access Signatures [Learn more](#)

Claims

☐

Manage

☐

Send

☒

Listen

Primary key

.....  

Secondary key

.....  

Primary connection string

.....  

Secondary connection string

.....  

SAS policy ARM ID

/subscriptions/d6549a66-c45c-4979-840c-3b356da446b0/resourceGroups/NewRG/... 

☐

Show AMQP connection strings

16. Then you need to open the Receive messages program which we used in the previous lab and paste the connection string there and run the program.
17. We are using this program to receive the log information from the event hub and you can see the information in place.
18. Also these are request that have logged by the azure web app and sent onto the event hubs.

```
D:\Azure\Azure DP-203\Desig x + -
Partition ID 0
Data Offset 0
Sequence Number 0
Partition Key
{"records": [{"time": "2025-03-12T05:27:52.1261293Z", "resourceId": "/SUBSCRIPTIONS/D6549A66-C45C-4979-840C-3B356DA446B0/RESOURCEGROUPS/NEWRG/PROVIDERS/MICROSOFT.WEB/SITES/NEWWEBAPP121", "category": "AppServiceHTTPLogs", "properties": {"CsMethod": "GET", "CsUriStem": "/", "SPort": "80", "CIp": "172.17.32.9", "UserAgent": "AlwaysOn", "CsHost": "newwebapp121-ajetabf3gjecdjdz.northeurope-01.azurewebsites.net", "ScStatus": 200, "ScSubStatus": "0", "ScWin32Status": "0", "ScBytes": 3242, "CsBytes": 487, "TimeTaken": 0, "Result": "Success", "Cookie": "-", "CsUriQuery": "-", "CsUsername": "-", "Referer": "-", "ComputerName": "dw0sdwk0001QM"}}, {"time": "2025-03-12T05:27:52.1265136Z", "resourceId": "/SUBSCRIPTIONS/D6549A66-C45C-4979-840C-3B356DA446B0/RESOURCEGROUPS/NEWRG/PROVIDERS/MICROSOFT.WEB/SITES/NEWWEBAPP121", "category": "AppServiceHTTPLogs", "properties": {"CsMethod": "GET", "CsUriStem": "/", "SPort": "443", "CIp": "127.0.0.1", "UserAgent": "AlwaysOn", "CsHost": "newwebapp121-ajetabf3gjecdjdz.northeurope-01.azurewebsites.net", "ScStatus": 200, "ScSubStatus": "0", "ScWin32Status": "0", "ScBytes": 3725, "CsBytes": 952, "TimeTaken": 12, "Result": "Success", "Cookie": "-", "CsUriQuery": "X-ARR-LOG-ID=f94688a0-2e8c-4233-946d-752acce67c9", "CsUsername": "-", "Referer": "-", "ComputerName": "dw0sdwk0001QM"}}, {"time": "2025-03-12T05:27:52.1342632Z", "resourceId": "/SUBSCRIPTIONS/D6549A66-C45C-4979-840C-3B356DA446B0/RESOURCEGROUPS/NEWRG/PROVIDERS/MICROSOFT.WEB/SITES/NEWWEBAPP121", "category": "AppServiceHTTPLogs", "properties": {"CsMethod": "GET", "CsUriStem": "/", "SPort": "80", "CIp": "172.17.32.9", "UserAgent": "AlwaysOn", "CsHost": "newwebapp121-ajetabf3gjecdjdz.northeurope-01.azurewebsites.net", "ScStatus": 200, "ScSubStatus": "0", "ScWin32Status": "0", "ScBytes": 3242, "CsBytes": 487, "TimeTaken": 19, "Result": "Success", "Cookie": "-", "CsUriQuery": "-", "CsUsername": "-", "Referer": "-", "ComputerName": "dw0sdwk0001QM"}}, {"time": "2025-03-12T05:27:52.1344384Z", "resourceId": "/SUBSCRIPTIONS/D6549A66-C45C-4979-840C-3B356DA446B0/RESOURCEGROUPS/NEWRG/PROVIDERS/MICROSOFT.WEB/SITES/NEWWEBAPP121", "category": "AppServiceHTTPLogs", "properties": {"CsMethod": "GET", "CsUriStem": "/", "SPort": "443", "CIp": "127.0.0.1", "UserAgent": "AlwaysOn", "CsHost": "newwebapp121-ajetabf3gjecdjdz.northeurope-01.azurewebsites.net", "ScStatus": 200, "ScSubStatus": "0", "ScWin32Status": "0", "ScBytes": 3725, "CsBytes": 952, "TimeTaken": 19, "Result": "Success", "Cookie": "-", "CsUriQuery": "X-ARR-LOG-ID=761ff367-0295-40e0-a4da-f2fe7dd37189", "CsUsername": "-", "Referer": "-", "ComputerName": "dw0sdwk0001QM"}]}]}
Partition ID 0
Data Offset 2568
Sequence Number 1
Partition Key
{"records": [{"time": "2025-03-12T05:28:52.0594174Z", "resourceId": "/SUBSCRIPTIONS/D6549A66-C45C-4979-840C-3B356DA446B0/RESOURCEGROUPS/NEWRG/PROVIDERS/MICROSOFT.WEB/SITES/NEWWEBAPP121", "category": "AppServiceHTTPLogs", "properties": {"CsMethod": "GET", "CsUriStem": "/", "SPort": "80", "CIp": "172.17.32.9", "UserAgent": "AlwaysOn", "CsHost": "newwebapp121-ajetabf3gjecdjdz.northeurope-01.azurewebsites.net", "ScStatus": 200, "ScSubStatus": "0", "ScWin32Status": "0", "ScBytes": 3242, "CsBytes": 487, "TimeTaken": 0, "Result": "Success", "Cookie": "-", "CsUriQuery": "-", "CsUsername": "-", "Referer": "-", "ComputerName": "dw0sdwk0001QM"}]}]}
```

19. After completing the lab do not delete the resources because we will use them in another lab.